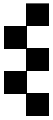
Multivalued Dependencies (MVD) and Update Anomalies



University of Essex

Multivalued Dependency

A multivalued dependency occurs when a determinant is matched with a particular set of values.

For example

The multivalued dependency

Person →→ Sibling ***(read as Person multidetermines Sibling)***

means that a person may have more than one sibling.

- *This is not a problem if the multivalued dependency exists in a separate table.*
- *But can cause a problem if the multivalued dependency exists within a table with other attributes.*

Example of Multivalued Dependency (1)

The determinant is associated with a set of values.

Employee Jones, for example, has degrees AA and BA. Employee Greene has degrees BS, MS, and PhD. Employee Chau has just one degree, BS.

EMPLOYEE_DEGREE

EmployeeName	EmployeeDegree
Chau	BS
Green	BS
Green	MS
Green	PhD
Jones	AA
Jones	BA

Examples of Multivalued Dependency (2)

The determinant is associated with a set of values.

Similarly, employee Jones has siblings (brothers and sisters) Fred, Sally, and Frank. Employee Greene has a sibling Nikki, and employee Chau has siblings Jonathan and Eileen.

EMPLOYEE_SIBLING

EmployeeName	EmployeeSibling
Chau	Eileen
Chau	Jonathan
Green	Nikki
Jones	Frank
Jones	Fred
Jones	Sally

Example of Multivalued Dependency (3)

The determinant is associated with a set of values.

For Example

PartKitName Bike Repair has parts Wrench, Screwdriver, and Tube Fix.

PARTKIT_PART

PartKitName	Part
Bike Repair	Screwdriver
Bike Repair	Tube Fix
Bike Repair	Wrench
First Aid	Aspirin
First Aid	Bandaids
First Aid	Elastic Band
First Aid	Ibuprofin
Toolbox	Drill
Toolbox	Drill bits
Toolbox	Hammer
Toolbox	Saw
Toolbox	Screwdriver

Reason for Multivalued Dependency

The possible existence of multi-valued dependencies in a relation is **due to 1NF**, which disallows an attribute in a tuple from having a set of values.

For example, if we have two multi-valued attributes in a relation, we have to **repeat each value of one** of the attributes **with every value of the other** attribute, to ensure that tuples of the relation are consistent.

Trivial and Nontrivial MVDs

- A MVD A −>> B in relation R is defined as being trivial if
  - (a) B is a subset of A

  or
  - (b) A union B = R.

  (when MVDs involves all attributes of a relation)

- A MVD is defined as being nontrivial if neither (a) nor (b) are satisfied.

Complementation Rule for MVDs

If $X \twoheadrightarrow Y$ holds in R, then $X \twoheadrightarrow R - (X \cup Y)$ also holds

For example,

For a relation R(A,B,C, D), If A ->> B holds

then A->> CD also holds.

However, for each fixed value of A, the possible values of B are independent of the remaining attributes.

Complementation Rule for MVDs (cont..)

For example, suppose $A = a_1$, and we have:

- $B \in \{b_1, b_2\}$
- $(C, D) \in \{(c_1, d_1), (c_2, d_2)\}$

If

$$A \twoheadrightarrow B$$

holds, then all combinations should be possible:

$$(a_1, b_1, c_1, d_1)$$

$$(a_1, b_1, c_2, d_2)$$

$$(a_1, b_2, c_1, d_1)$$

$$(a_1, b_2, c_2, d_2)$$

So B can vary without restricting which (C, D) values occur.

MVDs in a Relation: Example 1

BranchStaffOwner

branchNo	sName	oName
B003	Ann Beech	Carol Farrel
B003	David Ford	Carol Farrel
B003	Ann Beech	Tina Murphy
B003	David Ford	Tina Murphy

sName represents **staff name**
oName represents **property owner**

The following **nontrivial** MVDs exist in the relation:

branchNo →→ sName

branchNo →→ oName

Note: *sName and oName* are independent to each other.

MVDs in a Relation: Example 2

s_id	course	hobby
1	Science	Cricket
1	Maths	Hockey
2	C#	Cricket
2	Php	Hockey



s_id	course	hobby
1	Science	Cricket
1	Maths	Hockey
1	Science	Hockey
1	Maths	Cricket
2	C#	Cricket
2	PHP	Hockey
2	C#	Hockey
2	PHP	Cricket

The following **nontrivial** MVDs exist in the relation:

s_id →→ course

s_id →→ hobby

Note: *course* and *hobby* are independent to each other.

MVDs in a Relation: Definition

A relation is said to have a MVD, if the following conditions are true,

- if A →→ B in a relation, then the relation may have a MVD.
- relation should have at-least 3 attributes for it to have a MVD.
- for a relation R(A,B,C), if there is A →→ B , then B and C should be independent of each other.

If all these conditions are true for any relation, it will have a MVD.

MVDs in a Relation: Example 3

Name	Sibling	ShoeSize	Hobby
Chau	Eileen	C	Scuba diving
Chau	Jonathan	C	Scuba Diving
Green	Nikki	B	Embroidery
Jones	Frank	E	Model boats
Jones	Frank	E	Scuba diving
Jones	Fred	E	Model boats
Jones	Fred	E	Scuba diving
Jones	Sally	E	Model boats
Jones	Sally	E	Scuba diving

Name→→Sibling

Name→→Hobby

Name→ShoeSize

Insertion Anomaly due to MVDs

Insertion anomaly: If a Name adds a new Hobby, a line must be inserted for each Sibling.

For example, if Chau takes up the Hobby of model boats, two lines will have to be added – one for Sibling Jonathan and one for Sibling Eileen.

Name	Sibling	ShoeSize	Hobby
Chau	Eileen	C	Scuba diving
Chau	Jonathan	C	Scuba Diving
Green	Nikki	B	Embroidery
Jones	Frank	E	Model boats
Jones	Frank	E	Scuba diving
Jones	Fred	E	Model boats
Jones	Fred	E	Scuba diving
Jones	Sally	E	Model boats
Jones	Sally	E	Scuba diving

Name→→Sibling

Name→→Hobby

Name→ShoeSize

Deletion Anomaly due to MVDs

Deletion anomaly: If a Name drops a Hobby, one line must be deleted for each Sibling.

For example, if Jones drops the Hobby of model boats, three lines will have to be deleted – one for Sibling Fred, one for Sibling Sally and one for Sibling Frank.

Name	Sibling	ShoeSize	Hobby
Chau	Eileen	C	Scuba diving
Chau	Jonathan	C	Scuba Diving
Green	Nikki	B	Embroidery
Jones	Frank	E	Model boats
Jones	Frank	E	Scuba diving
Jones	Fred	E	Model boats
Jones	Fred	E	Scuba diving
Jones	Sally	E	Model boats
Jones	Sally	E	Scuba diving

Name→→Sibling

Name→→Hobby

Name→ShoeSize

Modification Anomaly due to MVDs

Modification anomaly: If a Hobby is changed for a Name with multiple siblings, we will have data integrity problems if the Hobby is not changed in every associated row.

For example, if Jones switches from model boats to model airplanes, we will have to modify three rows.

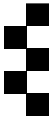
Name	Sibling	ShoeSize	Hobby
Chau	Eileen	C	Scuba diving
Chau	Jonathan	C	Scuba Diving
Green	Nikki	B	Embroidery
Jones	Frank	E	Model boats
Jones	Frank	E	Scuba diving
Jones	Fred	E	Model boats
Jones	Fred	E	Scuba diving
Jones	Sally	E	Model boats
Jones	Sally	E	Scuba diving

Name→→Sibling

Name→→Hobby

Name→ShoeSize

Fourth Normal Form (4NF)



University of Essex

Fourth Normal Form (4NF)

Any relation is said to be in the fourth normal form when it satisfies the following conditions :

- It must be in Boyce Codd Normal Form (BCNF).
- It should have no non-trivial multivalued dependency.

OR

- A relation is in 4NF if and only if for every nontrivial multi-valued dependency A —>> B, A is a candidate key of the relation.

Relation in 4NF: Example 1

BranchStaffOwner

branchNo	sName	oName
B003	Ann Beech	Carol Farrel
B003	David Ford	Carol Farrel
B003	Ann Beech	Tina Murphy
B003	David Ford	Tina Murphy

sName represents **staff name**
oName represents **property owner**

BranchStaff

branchNo	sName
B003	Ann Beech
B003	David Ford

BranchOwner

branchNo	oName
B003	Carol Farrel
B003	Tina Murphy

Update Anomaly Removed

- Both new relations are in 4NF as they contain only trivial MVDs.
- Relations in 4NF do not display data redundancy, and the potential for update anomalies is removed.
- **For example**, to **add a new property** owner for branch B003, we simply create a single tuple in the BranchOwner relation.

Relation in 4NF: Example 2

s_id	course	hobby
1	Science	Cricket
1	Maths	Hockey
2	C#	Cricket
2	Php	Hockey



s_id	course	hobby
1	Science	Cricket
1	Maths	Hockey
1	Science	Hockey
1	Maths	Cricket
2	C#	Cricket
2	PHP	Hockey
2	C#	Hockey
2	PHP	Cricket

s_id	course
1	Science
1	Maths
2	C#
2	Php

and

s_id	hobby
1	Cricket
1	Hockey
2	Cricket
2	Hockey

University of Essex

Properties of Normal Forms and their Decompositions

Property	3NF	BCNF	4NF
Eliminates redundancy due to FD's	No	Yes	Yes
Eliminates redundancy due to MVD's	No	No	Yes
Preserves FD's	Yes	No	No
Preserves MVD's	No	No	No

Advantages and disadvantages of Normalisation

Normalized tables have the advantages of

(1) eliminating update anomalies

(2) reducing data duplication.

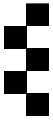
Disadvantages of normalized tables are

(1) more complicated SQL may be required for multitable subqueries and joins

and

(2) the extra work the DBMS has to do to run these SQL statements may result in slower application performance.

Fifth Normal Form (5NF)



University of Essex

Fifth Normal Form (5NF)

- Sometime **lossless decomposition** of a relation **R** into **two sub-relations** is not possible.
- But there may be a lossless decomposition into **more than two sub-relations**.
- In addition, there may be **no functional dependency** in **R** that violates any normal form up to BCNF, and there may be **no nontrivial MVD** present in **R** either that violates 4NF. *In other words, R should be in 4NF*.
- We then resort to another dependency called the **join dependency** and, if it is present, carry out a multiway decomposition into fifth normal form (5NF).

Join Dependency (JD)

- A join dependency (JD) says that a relation can be decomposed into several smaller relations and then reconstructed exactly by joining them back together.

- A join dependency (JD) can be said to exist if the join of **R1**(A,B), **R2**(A,C) and **R3**(B,C) is equal to relation **R** (A, B, C).

Fifth Normal Form (5NF)

Any relation R is said to be in the fifth normal form when it satisfies the following conditions :

- Relation R must be in fourth normal form (4NF)
- and further lossless decomposition of relation R is not possible

Example: Relation in 5NF

(b) **PropertyItemSupplier** (Legal state)

propertyNo	itemDescription	supplierNo
PG4	Bed	S1
PG4	Chair	S2
PG16	Bed	S2
PG4	Bed	S2

Decomposition in 5NF

PropertyItem

propertyNo	itemDescription
PG4	Bed
PG4	Chair
PG16	Bed

ItemSupplier

itemDescription	supplierNo
Bed	S1
Chair	S2
Bed	S2

PropertySupplier

propertyNo	supplierNo
PG4	S1
PG4	S2
PG16	S2

Another Definition of 5NF

- Discovering JDs in practical databases with hundreds of attributes is next to impossible.
- It can be done only with a great degree of intuition about the data on the part of the designer.
- Therefore, the current practice of database design pays scant attention to them.
- One result due to Date and Fagin (1992) relates to conditions detected using FDs alone and ignores JDs completely.
    - "If a relation/schema is in 3NF and each of its keys consists of a single attribute, it is also in 5NF."

Suggested Readings

- https://opentextbc.ca/dbdesign01/chapter/chapter-12-normalization/
- https://www.geeksforgeeks.org/normal-forms-in-dbms/?ref=lbp
- *Database Systems: A Practical Approach to Design, Implementation, and Management* by Connolly and Begg, Global Edition/6th Edition.
  - 15.3, 15.4, 15.5, and 7.2.4
- *Database processing: fundamentals, design, and implementation* by David M. Kroenke; David J. Aue,14th Edition (Global Edition).
  - Section Normal Forms (in Chapter 3), page 182